Ting-Han (Timmy) Lin

PhD Student in Computer Science | University of Chicago

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Research Summary

I design and evaluate socially intelligent robots that build long-term relationships with people. My work combines conversational robot systems, studies of group dynamics, longitudinal deployments with adults and children, and validated measures of human-robot rapport.

Education

University of Chicago | Chicago, IL

PhD in Computer Science | Sep 2023 – Jun 2028 (expected)

Advisor: Prof. Sarah Sebo

GPA: 4.00/4.00

Relevant Coursework**:** Robotics**,** Deep Learning Systems, Computational Linguistics, Behavioral Experiments for Human-Computer Interaction

University of Chicago | Chicago, IL

MS in Computer Science and BA in Economics | Sep 2018 – Jun 2022

Four-year joint BA/MS program

Advisors: Prof. Sarah Sebo and Prof. Ken Nakagaki

GPA: 3.77/4.00

Technical Skills

Programming: Python, R, Java, MATLAB, Kotlin, JavaScript, SQL, C#, C, HTML, CSS

Machine learning and AI: PyTorch, neural network training, image classification, XGBoost, large language model (LLM) API integration, prompt engineering, vision-language models, speech-to-text, text-to-speech

Robotics and prototyping: Robot Operating System (ROS), Android development, robot SDK integration, closed-loop control, RGB-D tracking, camera calibration, OpenCV, PyQt, Processing, Unity 3D, Arduino, Fusion 360, MoveIt, Gazebo

Robotics algorithms (teaching): A* path planning, particle-filter localization, Q-learning, forward and inverse kinematics

Data analysis: exploratory and confirmatory factor analysis (EFA/CFA), scale validation, linear and generalized linear mixed-effects models, analysis of variance (ANOVA), LIWC, Qualtrics, Azure Synapse Analytics, Apache Spark, Databricks, Tableau

Software development and tools: Django, Cython, Docker, Git

Research Experience

UChicago Department of Computer Science | Chicago, IL

Doctoral Researcher, Human-Robot Interaction Lab | Sep 2023 – Present

Advisor: Prof. Sarah Sebo

- Designed and implemented a Kotlin-based Temi V3 Android application integrating speech recognition, LLM dialogue, text-to-speech, navigation, and expressive behaviors. Led a three-session study, finding that robot empathy improved rapport and perceived warmth.

- Built an autonomous family assistant robot with LLM-based homework support, social check-ins, GPT-4.1 vision-language reminders, on-device face recognition, and scheduled tasks coordinated through a Django web application server. Led a two-week deployment, observing sustained robot-initiated interactions and no significant change in reported relationship measures.
- Developed and validated the 18-item Connection-Coordination Rapport (CCR) Scale using exploratory and confirmatory factor analysis, reliability assessment, and convergent validity testing (HRI 2025, C5). Extended the scale to an eight-item version using expert-rated content validity, comparison of 35 candidate item combinations, and validation across independent samples (THRI 2026, J2).
- Implemented autonomous Python-controlled NAO interactions with Whisper transcription, GPT-3.5 response summarization, and coordinated verbal and nonverbal behaviors. Evaluated how robot agreement and disagreement shaped interpersonal closeness, identifying effects that depended on robot appearance and communication style (CHB, online 2025, J1).

UChicago Department of Computer Science | Chicago, IL

Research Assistant, Actuated Experience Lab | Jan 2022 – Aug 2023

Advisor: Prof. Ken Nakagaki

- Designed and programmed ThrowIO spatial interaction prototypes in Processing, using Kinect RGB-D tracking and closed-loop control of Sony toio robots to manipulate thrown objects on an overhanging surface (CHI 2023, C3).
- Integrated camera calibration, coordinate mapping, and trajectory estimation with robot positioning and coordinated push and drop behaviors. Built additional spatial interaction prototypes using physics simulation and projection mapping.
- Built Arduino electronics, soldered circuits, designed 3D-printed objects in Fusion 360, and fabricated components using waterjet and laser cutters for prototyping.

UChicago Department of Computer Science | Chicago, IL

Research Assistant, Human-Robot Interaction Lab | Sep 2020 – Aug 2023

Advisor: Prof. Sarah Sebo

- Built an immersive puzzle game with a Python Wizard-of-Oz interface, Vector Python SDK, and ROS speech and motor control. Evaluated robot and human game guides in a 42-participant study, finding that participants felt less judged and more connected with the robot (RO-MAN 2022, C1).
- Designed and evaluated robot-facilitated conversations between strangers using Wizard-of-Oz control of NAO. Analyzed conversation transcripts with LIWC and assessed participant and pair-level outcomes using mixed-effects models and ANOVA (CHI 2023, C2).

Publications

Journal Articles

(J2) The Reduced-Length Connection-Coordination Rapport (CCR) Scale.
Ting-Han Lin, Guan Chen, Bilge Mutlu, J. Gregory Trafton, Sarah Sebo
ACM Transactions on Human-Robot Interaction (THRI), 15(3), Article 57, 1–29, 2026.

(J1) The Impact of a Robot's Agreement (or Disagreement) on Human-Human Interpersonal Closeness in a Two-Person Decision-Making Task.
Ting-Han Lin, Yuval Rubin Kopelman, Madeline Busse, Sarah Sebo*, Hadas Erel* (*equal contribution)
Computers in Human Behavior (CHB), 174 (2026), 108807. Published online September 2025.

Full Conference Papers

(C5) Connection-Coordination Rapport (CCR) Scale: A Dual-Factor Scale to Measure Human-Robot Rapport.
Ting-Han Lin, Hannah Dinner, Tsz Long Leung, Bilge Mutlu, J. Gregory Trafton, Sarah Sebo
In Proc. ACM/IEEE HRI 2025. Melbourne, Australia.

(C4) Role-Playing with Robot Characters: Increasing User Engagement through Narrative and Gameplay Agency.
Spencer Ng, **Ting-Han Lin**, You Li, Sarah Sebo
In Proc. ACM/IEEE HRI 2024. Boulder, USA.

(C3) ThrowIO: Actuated TUIs that Facilitate "Throwing and Catching" Spatial Interaction with Overhanging Mobile Wheeled Robots.
Ting-Han Lin, Willa Yunqi Yang, Ken Nakagaki
In Proc. ACM CHI 2023. Hamburg, Germany.

(C2) Ice-Breaking Technology: Robots and Computers Can Foster Meaningful Connections between Strangers through In-Person Conversations.
Alex Wuqi Zhang, **Ting-Han Lin**, Xuan Zhao, Sarah Sebo
In Proc. ACM CHI 2023. Hamburg, Germany.

(C1) Benefits of an Interactive Robot Character in Immersive Puzzle Games.
Ting-Han Lin*, Spencer Ng*, Sarah Sebo (*equal contribution)
In Proc. IEEE RO-MAN 2022. Naples, Italy.

Short Conference Papers

(S1) Understanding the Formation of Human-Robot Rapport to Facilitate Positive Long-Term Human-Robot Interaction.
Ting-Han Lin
In Proc. ACM/IEEE HRI 2026. Edinburgh, UK.

Demonstrations

(D1) Joint Demonstration of ThrowIO and AeroRigUI
ThrowIO: **Ting-Han Lin**, Willa Yunqi Yang, Ken Nakagaki
AeroRigUI: Lilith Yu*, Chenfeng Gao*, David Wu, Ken Nakagaki (*equal contribution)
In Proc. ACM CHI 2023 Interactivity Demo. Hamburg, Germany.

Industry Experience

Upfront Healthcare Services | Chicago, IL

Data Science Fellow | Jun 2021 – Feb 2022

- Developed and automated anomaly-detection data pipelines using Azure Synapse Analytics and Apache Spark, and Databricks.
- Built SQL reports and Tableau dashboards to communicate data insights.

Actelligent Capital | Hong Kong

Data Scientist Intern | Jun 2020 – Sep 2020

- Used SQL and Python to process more than 20 years of company financial data and calculate Fama-French factor metrics.
- Developed XGBoost predictive models using financial indicators to assess stock performance and risk.
- Evaluated Cython optimizations for financial metric calculations and presented performance recommendations.

Shanghai Commercial & Savings Bank (SCSB) | Taipei, Taiwan

Financial Services Intern | Jun 2019 – Aug 2019

- Analyzed and improved the marketing strategy of SCSB's credit card to increase its reach, engagement, and sales through Google Analytics and AdWords.
- Used Python to extract database records and analyze audience engagement with online banking.

Honors and Awards

Daniels Fellowship, University of Chicago, 2023–2024

$5,000 award for selected incoming doctoral students.

Dean's List, University of Chicago, 2018–2019, 2021–2022

Recognized for a GPA in the top 20% for the academic year.

Bridge Builder Award, University of Chicago, 2021

Recognized for building connections between student groups and individuals.

Metcalf Fellowship Grant, University of Chicago, 2019, 2020, 2021

$5,000 per award supporting internship experiences.

Teaching Experience

Teaching Assistant in 14 appointments across 8 courses, University of Chicago

Responsibilities across appointments included delivering lectures, leading labs, discussions, and office hours, preparing course materials, and assessing assignments, projects, and exams.

Human-Robot Interaction: Research and Practice (CMSC 20630/CMSC 30630)

Spring 2026 | Instructor: Prof. Sarah Sebo | Enrollment: 20

Spring 2025 | Instructor: Prof. Sarah Sebo | Enrollment: 20

- Authored an LLM-based human-robot dialogue lab integrating Rev.ai or OpenAI Whisper speech-to-text, Gemini multi-turn conversation, OpenAI text-to-speech, prompt engineering, and expressive robot behaviors.
- Authored a Wizard-of-Oz lab guiding students to build Python GUIs for real-time control of Misty speech, gestures, facial displays, and LEDs.

Introduction to Robotics (CMSC 20600/CMSC 30600)

Spring 2024 | Instructor: Prof. Sarah Sebo | Enrollment: 27

- Developed a PyTorch image-classification lab, covering neural network training and ROS integration of camera-based inference to control TurtleBot3 behaviors.
- Supported robotics instruction and student project debugging in ROS, covering A* path planning, particle-filter localization, Q-learning, forward and inverse kinematics, and robot-arm control with MoveIt and Gazebo.

Mathematical Methods for Data Science I (DATA 21100)

Winter 2024 | Instructor: Prof. David Biron | Enrollment: 28

Winter 2023 | Instructors: Prof. David Biron, Prof. Amy Nussbaum | Enrollment: 54

- Taught mathematical foundations for machine learning, including probability distributions, conditional probability, multivariable calculus, gradients, and gradient descent.

User Interface and User Experience Design (MPCS 51230)

Autumn 2023 | Instructor: Prof. André Marques | Enrollment: 30

Winter 2023 | Instructor: Prof. André Marques | Enrollment: 37

- Taught Figma prototyping, including plugin use, interactive user flows, and high-fidelity interface design, alongside human-centered design methods such as card sorting and usability testing.

Mathematical Methods for Data Science II (DATA 21200)

Spring 2023 | Instructor: Prof. David Biron | Enrollment: 49

Spring 2022 | Instructor: Prof. David Biron | Enrollment: 27

- Taught linear algebra methods for machine learning, including orthogonal projections, eigendecomposition, LU decomposition, and singular value decomposition (SVD).

Introduction to Human-Computer Interaction (CMSC 20300/MAAD 25300)

Autumn 2022 | Instructors: Prof. Pedro Lopes, Prof. Ken Nakagaki | Enrollment: 93

- Supported student projects in VR interaction design and web development, providing technical guidance on prototyping, user experience evaluation, HTML5, CSS3, JavaScript, and PHP server-side programming.

Introduction to Data Science I (CMSC/STAT/DATA 11800)

Winter 2022 | Instructor: Prof. Will Trimble | Enrollment: 81

Autumn 2021 | Instructors: Prof. Michael Franklin, Prof. Dan Nicolae | Enrollment: 90

Autumn 2020 | Instructors: Prof. Michael Franklin, Prof. Dan Nicolae | Enrollment: 94

- Taught data analysis in Python and Jupyter Notebook, guiding students through data collection, storage and processing, exploratory analysis, visualization, statistical inference, prediction, and interpretation of results.

Introduction to Data Science II (CMSC/STAT 11900)

Winter 2021 | Instructors: Prof. Michael Franklin, Prof. Dan Nicolae | Enrollment: 67

- Assessed students' application of probability, statistical methods, and basic machine learning to real-world problems, with attention to data privacy, ethics, encryption, and reproducibility.

Mentoring

Mentored 24 undergraduate, master's, and high school students in research design, robot programming, and data analysis. Publication identifiers below indicate associated coauthored papers.

Sam Wang, UChicago Undergraduate Student | Autumn 2026 – Present

Xinyi Jiang, UChicago Master's Student | Autumn 2026 – Present

Rey Kim, UChicago Master's Student | Spring 2026 – Summer 2026

Elaine Liang, UChicago Master's Student | Winter 2026 – Present

Ricky Bao, UChicago Undergraduate Student | Autumn 2025 – Summer 2026

Uma Malani, UChicago Undergraduate Student | Summer 2025 – Summer 2026

Sabrina Fajardo, UChicago Undergraduate Student | Summer 2025 – Summer 2026

Awa Cisse, Carleton College Undergraduate Student | Summer 2026

Arjun Chopra, UChicago Undergraduate Student | Summer 2026

Jingyu Chen, UChicago Master's Student | Autumn 2025 – Spring 2026

Guan Chen (J2), UChicago Undergraduate Student | Winter 2024 – Winter 2026

Rachel Liu, UChicago Undergraduate Student | Spring 2024 – Spring 2025

Tsz Long Leung (C5), UChicago Undergraduate Student | Spring 2024 – Spring 2025

Ike Peng, UChicago Undergraduate Student | Winter 2024 – Spring 2025

Madeline Busse (J1), UChicago Undergraduate Student | Summer 2023 – Spring 2025

Hannah Dinner (C5), High School Summer Intern at UChicago | Summer 2024 – Autumn 2024

Frida Xuewei Li, UChicago Master's Student | Spring 2024 – Autumn 2024

Aarman Pannu, UChicago Undergraduate Student | Winter 2024 – Spring 2024

Joy Chen, UChicago Master's Student | Autumn 2023 – Winter 2024

Clark Kovacs, UChicago Undergraduate Student | Autumn 2023 – Winter 2024

Liberto de Pablo, UChicago Undergraduate Student | Autumn 2023 – Winter 2024

Lucas Pardo, UChicago Undergraduate Student | Autumn 2023 – Winter 2024

You Li (C4), UChicago Undergraduate Student | Summer 2023 – Winter 2024

Esha Mujumdar, High School Summer Intern at UChicago | Summer 2023

UChicago Physical Sciences Division Peer Mentor | Autumn 2024 – Spring 2025

- Guided mentees in achieving personal, academic, and professional growth through one-on-one mentorship.
- Fostered an inclusive community by promoting interdisciplinary connections and a sense of belonging within the division.

UChicago Peer Mentor and Peer Mentor Advisory Board | Spring 2019 – Autumn 2021

- Led training sessions for more than 100 UChicago peer mentors.
- Supported incoming first-year and transfer students as they transitioned to the University of Chicago.

Professional Service

UChicago Department of Computer Science People and Technology (PAT) Seminar Series

Organizer for Department Seminar Series

- Organized student and faculty talks for the PAT Seminar Series (Spring 2026, Autumn 2026).

Computers in Human Behavior

Paper Reviewer

- Reviewed 1 paper for CHB (2026).

ACM Symposium on User Interface Software and Technology (UIST)

Paper Reviewer

- Reviewed 2 papers for UIST (2026), receiving one Special Recognition for Outstanding Reviews.

ACM Conference on Tangible, Embedded, and Embodied Interaction (TEI)

Student Volunteer

- Assisted with in-person conference operations and sessions at TEI 2026.

ACM Transactions on Human-Robot Interaction (THRI)

Paper Reviewer

- Reviewed 5 papers for THRI (2025, 2026).

International Journal of Human-Computer Interaction (IJHCI)

Paper Reviewer

- Reviewed 3 papers for IJHCI (2026).

ACM/IEEE Human-Robot Interaction (HRI)

Program Chair, Pioneers Workshop

- Coordinate submission reviews, reviewers, and the workshop program for HRI 2027.

Student Volunteer

- Assisted with in-person conference operations and sessions at HRI 2026.

Paper Reviewer

- Reviewed 12 papers for HRI (2024, 2025, 2026, 2027), receiving 2 Special Recognitions for Outstanding Reviews.

ACM Conference on Human Factors in Computing Systems (CHI)

Paper Reviewer

- Reviewed 6 papers for CHI (2024, 2025, 2026, 2027).

Augmented Humans (AHs)

Paper Reviewer

- Reviewed 1 paper for AHs (2024).

Outreach

Summer Robotics Course for South Side Chicago High School Students | Chicago, IL

Pre-Collegiate Non-Degree Program Assistant and Teaching Assistant | Jun 2024 – Aug 2024

- Taught local South Side Chicago high school students how to design and program robots using Blockly, a block-based programming tool.

Robot Demos at Chicago's Museum of Science and Industry | Chicago, IL

Demonstrator | Apr 2022 – Apr 2025

- Demonstrated research and robots at Museum of Science and Industry's National Robotics Week Block Party exhibit.

Volunteering and Leadership

UChicago Physical Sciences Division Social Committee | Chicago, IL

President | Feb 2025 – Jun 2026

Secretary | Feb 2024 – Feb 2025

- Organized social, athletic, and cultural events for graduate students in the Physical Sciences Division, made funding allocation decisions, and maintained communication with the graduate student body.

FIRST LEGO League Robotics Tournament | Chicago, IL

Robot Design Judge and Table Referee | Oct 2021 – Dec 2022

- Evaluated robot performance, code, mission strategy, iterative testing, and students' explanations of their design process.

Be The Match (National Marrow Donor Program) | Chicago, IL

Volunteer | Sep 2019 – Jun 2022

- Organized annual donor registration events at the UChicago student center and recruited potential blood stem cell donors.

UChicago Intercollegiate Financial Journal (IFJ) | Chicago, IL

Senior Journal Writer | Oct 2018 – Jun 2020

- Published articles on advances in artificial intelligence for the undergraduate community.
- Pitched article ideas and guided junior writers in weekly seminars.

Certifications

Programming

Oracle Certified Associate Java Programmer (2018)

Finance

Futures Specialist Certification (2026)

Securities Specialist Certification (2019)

Negotiation

Certificate of Advanced Negotiation, The Wharton School (2017)

Digital Tools and Marketing

Microsoft Office Specialist 2010 Master (2018)

Adobe Certified Associate (2018)

Google Analytics Certification (2018)

Google AdWords Certification (2018)

Selected Press

UChicago CS News. “New Prototypes AeroRigUI and ThrowIO Take Spatial Interaction to New Heights – Literally” Apr 18, 2023

UChicago CS News. “UChicago Research Tests Whether Robots or Humans Are Better Game Partners” Oct 18, 2022